Project #2 – FightSim

Learning Objectives

- Apply object-oriented programming concepts in C++

Overview

In this project, you will build a combat simulator for a simplified roll playing game (RPG). You will create a C++ class for characters in the game, including the character's name, hit points, armor class, and so forth. You will then create two character objects and simulate a combat between them.

The Character Class

You should define a C++ class called Character that implements an RPG character in the files character.h and character.cpp.

Character Data

Your character class should contain the following private member data.

Name	Type	Description
`name`	`string`	Character's name
`role`	`string`	Type of character (e.g. wizard, fighter, rogue, etc.)
`hitPoints`	`Int`	Character's current health; When it reaches zero, the character is dead.
`attackBonus`	`int`	Value added to a character's attack roll to determine if an attack hits.
`damageBonus`	`int`	Value added to a character's damage roll when attacking.
`armorClass`	`int`	Value determines how hard a character is to hit

Character member functions

Your Character class should implement, minimally, the following member functions (methods):

Name	Return Type	Description
`print(ostream& os)`	`ostream&`	Print the character to the given ostream
`attack(Character& otherCharacter)`	`void`	Attack another character (see below for details)
`damage(int amount)`	`void`	Subtract amount from the character's current hit points (Note: the minimum hit point value should be zero, so you will need to check for negative results)
`getHealth()`	`int`	Return the character's current health
`getName`	`string`	Return the character's name
`getRole()`	`string`	Return the character's role

Attacking another character

You should implement a member function for attacking another character. To have character1 attack character2, for example, you would call character1.attack(character2); When the attack member function is called, the following steps should be taken:

1. Roll a 20-sided die (i.e. generate a random number from 1 to 20) and add the attacking character's attackBonus.
2. If the result is equal to or higher than the opponent's armorClass, then the attack hits. Otherwise it misses.
3. If the attack hits, roll a 10-sided die and add the attacking character's damageBonus. Subtract the resulting amount from the opponent's hit points by calling it's damage(amount) method.

Program Operation

Your main program should be implemented in a file project2.cpp. Your program should prompt the user for character details (name, role, hit points, etc.) for two characters. It should then simulate combat between the two characters until one of the two characters reaches zero hit points.

I suggest implementing the input prompt *after* the character and combat logic. You can simply hard-code values for each character, then when you test the program you will not have to enter values each time you run it.

Example of Program Execution

```
First character name?
Uglar

Uglar's role?
Barbarian

Uglar the Barbarian's hit points?
80

Uglar the Barbarian's attack bonus?
5

Uglar the Barbarian's damage bonus?
5

Uglar the Barbarian's armor class?
24

Character summary
-----------------
Uglar the Barbarian
HP: 80
AB: 5
DB: 5
AC: 24

Second character name?
Zimzizz

Zimzizz's role?
Wizard

Zimzizz the Wizard's hit points?
40

Zimzizz the Wizard's attack bonus?
5

Zimzizz the Wizard's damage bonus?
15

Zimzizz the Wizard's armor class?
18

Character summary
-----------------
Zimzizz the Wizard
HP: 40
AB: 5
DB: 15
AC: 18
```

Example of Program Execution (con't)

```
Simulated Combat:
Uglar attacks!
Attack roll: 14 + 5 = 19 --> HIT!
Damage: 9 + 5 = 14
Zimzizz has 26 hit points remaining

Zimzizz attacks!
Attack roll: 17 + 5 = 22 --> MISS!

Uglar attacks!
Attack roll: 13 + 5 = 18 --> HIT!
Damage: 8 + 5 = 13
Zimzizz has 13 hit points remaining

Zimzizz attacks!
Attack roll: 19 + 5 = 24 --> HIT!
Damage:  8 + 15 = 23
Uglar has 57 hit points remaining

Uglar attacks!
Attack roll: 16 + 5 = 21 --> HIT!
Damage: 9 + 5 = 14
Zimzizz has 0 hit points remaining

Uglar wins!
```

Turn in and Grading

Please zip your project2.cpp, character.h, and character.cpp files into a single archive, and upload the zip file to the Dropbox.

This project is worth 10 points, distributed as follows

Task	Points
Program compiles, runs, and produces correct output	4
The Character class is correctly defined in a .h and .cpp file as described above.	1
The Character class includes all member data and member functions described above	2
Program conforms to coding standards (see CS 3100 Coding Standards in the Content/Handouts section of our class Pilot page)	3